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Beta2-Receptor Agonists and Antagonists

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Continuing Education Activity

Beta-2 adrenergic receptors play a central role in the management of bronchospastic disorders and are among the most frequently targeted pathways in respiratory and cardiovascular care. This activity examines the molecular mechanisms of β2 receptor signaling and compares short-, long-, and ultra-long-acting agonists, as well as nonselective and cardioselective antagonists. This activity also reviews the indications, routes of administration, and evidence-based dosing strategies. Additionally, this activity identifies key adverse effects of β2 agonists and antagonists, and implements monitoring protocols for serum electrolytes, heart rate, and pulmonary function to ensure patient safety.

Furthermore, this activity also covers the management of toxicity, including recognition of overdose symptoms and the appropriate use of beta-blocker antidotes when indicated. This activity also emphasizes the importance of interprofessional collaboration among healthcare providers: pharmacists can focus on medication reconciliation and screening for potential drug interactions; nurses can gain expertise in inhaler education and monitoring vital signs; and physicians will be guided in applying guideline-driven therapy selection and managing acute exacerbations, particularly in patients with **asthma** and COPD.

Objectives:

- Identify the pharmacological mechanisms, receptor subtypes, and physiological effects of β2 adrenergic agonists and antagonists.
- Implement evidence-based dosing regimens and appropriate routes of administration for β2-targeted medications.
- Select the most appropriate β2 agonist or antagonist based on patient-specific factors, disease severity, and guideline recommendations.
- Collaborate with interprofessional healthcare providers to ensure safe and effective therapy and manage acute exacerbations and adjustment of treatment for patients with comorbid conditions.

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Indications

Beta (β)-adrenergic receptors are transmembrane glycoproteins that elicit intracellular responses upon binding to catecholamines. They belong to the G-protein–coupled receptor (GPCR) family, also known as R7G, and are specifically linked to guanine nucleotide (GTP)–binding proteins (G proteins).[1]

β-Adrenergic receptors are classified into the following 3 subtypes:

- β1 Receptors, which are located mainly in cardiac tissues.

- β2 Receptors, which are distributed widely across different organs.
- β3 Receptors, which are concentrated in adipose tissues and the urinary bladder.

Other adrenergic receptor classes include alpha-1 (α1) and alpha-2 (α2).

Among these, β2 receptors are expressed throughout the human body, although their density and functional effects vary by tissue type. They are most abundant in the bronchial smooth muscle of the lungs, particularly in the smaller airways of the lungs. High receptor levels are also found in the vascular smooth muscle of skeletal muscle beds and in the smooth muscles of the gastrointestinal tract, uterus, and urinary bladder, producing different physiological responses depending on their location. In addition, β2 receptors are also present in cardiac muscle, liver, pancreas, adipose tissue, and specific brain regions—particularly the cerebellum and hippocampus—as well as on immune cells such as eosinophils and lymphocytes.

Natural catecholamines, such as adrenaline and noradrenaline, exhibit a broad spectrum of activity by interacting with multiple adrenergic receptor subtypes, which can potentially lead to diverse physiological effects. Both endogenous and synthetic adrenergic ligands differ in their affinity for these receptor subtypes. As a result, some synthetic **drugs** are developed to selectively target specific adrenergic receptors or their subtypes, thereby enhancing therapeutic efficacy and minimizing unintended adverse effects.

β2 Adrenergic receptors are primarily activated by natural catecholamines, with epinephrine (also known as adrenaline) being the most potent endogenous agonist, and norepinephrine (also known as noradrenaline) producing a weaker effect on these receptors. Epinephrine-mediated β2 receptor activation elicits important physiological responses, including bronchodilation and vasodilation. To enhance therapeutic benefits and reduce adverse effects, synthetic β2 receptor agonists have been developed with improved receptor selectivity. Conversely, β2 receptor antagonists, commonly referred to as β-blockers, inhibit the actions of catecholamines on these receptor sites. Although nonselective β-blockers act on both β1- and β2 receptors, β2-selective antagonists are rarely used in clinical practice due to their limited therapeutic indications.

Beta-2 Adrenergic Receptor Agonists

β2 Adrenergic receptor agonists are **drugs** that selectively stimulate β2 adrenergic receptors. These receptors are classified as sympathomimetics, and they mimic the effects of endogenous catecholamines such as epinephrine and norepinephrine; however, their activity is largely confined to β2 receptors. Clinically, they are primarily used for their targeted action on airway smooth muscle, resulting in relaxation and bronchodilation.[2][3] These agents are categorized based on their duration of action, as mentioned below.

Short-acting β2 agonists: Short-acting β2 agonists (SABAs), including albuterol (also known as salbutamol), levalbuterol, metaproterenol, and terbutaline, are primarily prescribed for the relief of acute bronchospasm associated with conditions such as **asthma** and chronic obstructive pulmonary disease (COPD). Their rapid onset and short duration of action make them suitable for the relief of acute symptoms.[4][5]

Beyond their primary indications, albuterol is also utilized off-label for the management of hyperkalemia. In contrast, terbutaline has off-label applications in delaying preterm labor and treating peripheral ischemia caused by vasopressor extravasation, with variable responses.[6][7][8][9]

Long-acting beta-2 agonists: Long-acting β2 agonists (LABAs), including salmeterol, formoterol, and arformoterol, are indicated for the maintenance treatment of bronchoconstriction in patients with **asthma**, COPD, chronic bronchitis, and emphysema. Certain LABAs, particularly formoterol, may also be used for acute management of **asthma** in combination with inhaled corticosteroids.[10]

Ultra-long-acting beta-2 agonists (ULABAs): Ultra-long-acting β2 agonists (ULABAs), including olodaterol, vilanterol, and indacaterol, have proven their potential to provide sustained, once-daily bronchodilation for the

management of COPD. They were approved by the US Food and Drug Administration (FDA) for this indication. Vilanterol is also FDA-approved for use in the management of **asthma**.[11]

A hazardous off-label use of clenbuterol—a LABA that is not FDA-approved and is mainly used to treat respiratory disorders in horses—is its use by bodybuilders for presumed anabolic and lipolytic properties, for which there is no evidence of efficacy in humans.[12]

Beta-2 Adrenergic Receptor Antagonists (Beta Blockers)

β2 Adrenergic receptor antagonists, also known as β-blockers, inhibit the activation of β-adrenergic receptors. Nonselective β-blockers, such as propranolol, timolol, and carvedilol, block both β1- and β2 receptors, affecting heart rate, contractility, and bronchoconstriction. Notably, although β-blockers are widely used in cardiovascular conditions, their use in patients with **asthma** or COPD requires careful consideration due to potential bronchoconstrictive effects. Selective β2 antagonists, such as butoxamine, exist, but they are not FDA-approved for any clinical use.[13]

Mechanism of Action

β2 Adrenergic receptors are primarily encoded on chromosome 5 and are predominantly expressed on the smooth muscle cells of the airways. Structurally, β2 receptors belong to the GPCR family and consist of 7 transmembrane helices and a short intracellular helix (often called helix 8) that lies parallel to the membrane. The extracellular loops facilitate ligand binding, and the intracellular loops interact with G proteins.[14][15]

Upon activation by agonists, β2 receptors undergo a conformational change that enables them to couple with heterotrimeric Gs proteins, consisting of 3 subunits: alpha, beta, and gamma. The receptor exists in a dynamic equilibrium between active and inactive conformations, and agonists stabilize the active form. Activation of the β2 receptor leads to the exchange of GDP for GTP on the alpha subunit of the Gs protein. The GTP-bound alpha subunit then dissociates from the beta-gamma complex and activates adenylate cyclase. This enzyme catalyzes the conversion of adenosine triphosphate (ATP) into cyclic adenosine monophosphate (cAMP)—a key intracellular second messenger.

Elevated intracellular cAMP levels trigger a cascade of events leading to smooth muscle relaxation via 2 main pathways. First, cAMP binds to the regulatory subunits of protein kinase A, thereby releasing its catalytic subunits, which then phosphorylate various target proteins that mediate smooth muscle relaxation. The second mechanism involves reducing intracellular calcium concentration through protein kinase A–mediated phosphorylation, which inhibits calcium influx, decreases calcium release from intracellular stores, and enhances calcium sequestration, ultimately diminishing smooth muscle contraction.

This signaling mechanism not only promotes bronchodilation in the airways but also contributes to uterine relaxation, which is a property utilized therapeutically in tocolysis.[16] Additionally, β2 receptor activation influences several other physiological processes, including stimulating the breakdown of glycogen to increase blood glucose levels, enhancing mucociliary clearance by increasing the beating frequency of cilia, decreasing acetylcholine release to support bronchodilation, changing vascular permeability, and modulating immune cell function.[17][2][18]

In contrast, β-blockers inhibit these processes by stabilizing β2 receptors in an inactive conformation, thereby preventing their activation. Clinically, this antagonism can worsen conditions such as **asthma** by negating the bronchodilatory effects of endogenous catecholamines.[19]

Administration

Beta-2 Receptor Agonists

β2 Agonists are primarily administered by inhalation—via metered-dose inhalers (MDIs), dry powder inhalers (DPIs), or nebulized formulations—to maximize bronchodilation in the airways while minimizing systemic exposure and adverse effects (Global Initiative for **Asthma** (GINA) Strategy 2021: Executive Summary and Rationale for Key Changes). Inhaled delivery ensures high local drug concentrations with reduced doses, thereby effectively targeting the respiratory

system. In addition, fixed-dose combination inhalers containing both β2 agonists and inhaled corticosteroids help reduce systemic exposure to corticosteroids in **asthma** maintenance therapy.[20]

In severe, acute **asthma** exacerbations, where rapid systemic action is critical, β2 agonists may be administered intravenously (IV) or intramuscularly (IM). Although oral formulations of LABAs and ULABAs are available for the long-term management of **asthma** and COPD, inhalation remains the preferred route due to its similar efficacy and better safety profile (see **Table 1**).[21] Additionally, subcutaneous terbutaline has been used in **asthma** exacerbations, although supporting evidence remains limited.[22]

Abbreviations: DPI, dry powder inhaler; IM, intramuscular; IV, intravenous; MDI, metered-dose inhaler.

β2 Agonists, including both SABAs and LABAs, remain fundamental to the management of **asthma** and COPD. The Global Initiative for **Asthma** (**GINA**) and the Global Initiative for Chronic Obstructive Lung Disease (**GOLD**) guidelines provide comprehensive recommendations to help clinicians in selecting the appropriate type of β2 agonist based on individual patient needs and disease severity.[39][40]

Beta Blockers

β2 Antagonists (β-blockers), including propranolol, timolol, and carvedilol, are primarily administered orally or IV to achieve systemic cardiovascular effects, such as reduced heart rate, decreased myocardial contractility, and lowered blood pressure (see **Table 2**). Oral administration is generally preferred for the chronic management of conditions such as hypertension, arrhythmias, and heart failure, offering consistent plasma levels and convenient dosing. In acute settings—such as hypertensive emergencies, arrhythmia control, or thyrotoxic crisis—IV formulations of nonselective β-blockers, including propranolol, may be used for rapid onset of action.[41]

Timolol is uniquely available in ophthalmic formulations for the treatment of open-angle glaucoma, where it provides localized β-receptor blockade to reduce intraocular pressure while minimizing systemic absorption.[42][43] Although β2-selective antagonists such as butoxamine exist, they are not clinically approved for use and remain confined to research settings due to the absence of defined therapeutic applications.

The American Heart Association (AHA) (AHA/American College of Cardiology [ACC]/Heart Failure Society of America [HFSA] Guideline for the Management of Heart Failure) acknowledges the critical role of β-blockers in cardiovascular disease conditions but cautions that nonselective β-blockers (β2 antagonists) may be contraindicated in patients with reactive airway diseases.[46]

Adverse Effects

Beta-2 Receptor Agonists

At recommended doses, β2 agonists are generally well tolerated. However, higher or systemic doses can lead to adverse effects resembling sympathetic stimulation. The most common adverse event is tremor, with additional effects including palpitations and tachycardia.[47] Animal studies have shown that β-agonists may increase anxiety-like behavior, but human studies—especially those using inhaled β2 agonists in combination with corticosteroids—have not demonstrated a statistically significant increase in anxiety symptoms.[48][49]

At elevated doses, β2 agonists have also been linked to metabolic disturbances such as lactic acidosis, hypokalemia, hypomagnesemia, and hyperglycemia. While these effects are usually mild, they may become clinically significant in patients with diabetes, those receiving diuretics, or individuals undergoing repeated high-dose β2 agonist therapy (eg, in status asthmaticus).[47]

LABAs, which are used for long-term **asthma** control, may cause adverse effects such as cardiac dysrhythmias and muscle cramps, even at standard doses—particularly when prescribed without concurrent inhaled corticosteroids. Chronic LABA use has also been associated with tolerance, leading to a reduced bronchodilator response over time and

potentially worsening **asthma** control.[50][51] Rarely, β2 agonists may cause diastolic hypotension due to peripheral vasodilation.[52][53]

Observational studies have suggested a possible association between long-term β2 agonist use and an increased risk of cardiovascular events, including myocardial infarction, especially in patients with preexisting ischemic heart disease or when used without inhaled corticosteroids. However, other studies have not confirmed this association, highlighting the need for cautious interpretation.[54][55]

Some case reports have mentioned paradoxical bronchoconstriction as an adverse effect of β2 agonists, in which inhaled medication induces bronchoconstriction instead of the expected bronchodilatory effect.[56]

Beta-2 Receptor Antagonists (Beta Blockers)

β-Blockers with significant β2 receptor–blocking activity can also produce adverse effects. These may include Raynaud phenomenon due to enhanced peripheral vasoconstriction; exercise-induced hypoglycemia, as β2 stimulation supports glycogenolysis and gluconeogenesis; and, most importantly, increased airway resistance and muscle cramps.[18]

Contraindications

As with all medications, hypersensitivity to the active substance is a contraindication for both β2 receptor agonists and β-blockers.

β2 Agonists stimulate sympathetic activity, which may precipitate or worsen tachyarrhythmias and other cardiac conditions. Therefore, caution is advised in patients with known arrhythmias or significant cardiovascular disease.[54] β2 Agonists also promote potassium uptake into cells, potentially exacerbating existing hypokalemia. Although not an absolute contraindication, caution is warranted in patients with low baseline potassium levels.[47][53]

Notably, some DPI inhalers, which are most commonly used for albuterol administration, contain lactose; therefore, they are contraindicated in patients with hypersensitivity to lactose.[57] Finally, LABAs are contraindicated as monotherapy in **asthma**, as their use without concomitant corticosteroids has been linked to tolerance, increased adverse events, and **asthma**-related mortality.[58]

Nonselective β-blockers can cause bronchoconstriction and are generally contraindicated in patients with reactive airway diseases such as **asthma** and COPD. β-Blockers may also mask adrenergic symptoms of hypoglycemia (such as tachycardia and tremor) and interfere with normal glucose regulation, potentially leading to an increased risk of severe hypoglycemic episodes or hypoglycemic coma in diabetic patients.[18]

Cardioselective β-blockers are not absolutely contraindicated in patients with **asthma**; however, when β-blockers are required for cardiac indications in patients with **asthma** or other bronchospastic conditions, administration should be considered on a case-by-case basis. β1-selective agents (eg, metoprolol and bisoprolol) are preferred, as their relative selectivity minimizes the risk of bronchoconstriction. Treatment should be initiated under close supervision by a specialist.

Monitoring

The therapeutic index of β2 agonists varies among agents, and both their beneficial and adverse effects may differ between individuals. These differences are influenced by factors such as the severity of the underlying disease, the duration of therapy, and the use of concomitant medications.[47]

Plasma concentrations of β2 agonists correlate with the severity of toxicity, and while precise measurement using techniques such as gas chromatography, liquid chromatography coupled with mass spectrometry (LC–MS), or high-performance liquid chromatography (HPLC) with fluorimetric or electrochemical detection is possible, such assays are typically limited to research and anti-doping settings rather than routine clinical care.[59]

In clinical practice, efficacy and potential toxicity are primarily assessed by clinical parameters rather than direct measurements of drug levels. For patients receiving prolonged β2 agonist therapy, regular monitoring of serum potassium and blood glucose is recommended due to the risk of hypokalemia and hyperglycemia.[53]

Additionally, monitoring of heart rate and blood pressure is recommended to detect cardiovascular adverse effects early. Monitoring of β2 agonist levels is a part of anti-doping programs prior to athletic competitions, where these agents are monitored for their potential performance-enhancing effects.

Toxicity

β2 Agonist toxicity may occur both accidentally and intentionally. The most serious life-threatening complication is severe hypokalemia resulting from the intracellular shift of potassium, which predisposes patients to cardiac arrhythmias.[60] In addition, toxicity may also manifest as exaggerated forms of known adverse effects, including hypotension with a bounding pulse (predominantly affecting diastolic pressure), supraventricular tachycardia or ventricular extrasystoles, tremors, hyperglycemia, lactic acidosis, tachypnea, and pupillary dilation.[61][62] In cases of β2 agonist overdose, β-blockers such as propranolol and esmolol may be used as antidotes through competitive antagonism at β2 receptors.[63]

Toxic doses of β-blockers can exacerbate **asthma** and lead to serious complications, including bradycardia, prolonged QRS complexes, hypotension, arrhythmias, hypoglycemia, hypothermia, severe peripheral arterial disease, high-grade sinoatrial or atrioventricular block, and variant angina.[64][65][66]

Enhancing Healthcare Team Outcomes

Asthma and COPD are highly prevalent chronic respiratory conditions that significantly impact patients' quality of life worldwide. Although treatment guidelines vary across regions, β2 adrenergic agonists consistently remain a cornerstone of both acute and chronic management due to their potent bronchodilatory effects. Their ubiquitous role in therapy underscores the importance of clinicians maintaining a thorough and up-to-date understanding of their pharmacological properties, including mechanisms of action, potential adverse effects, contraindications, and concerns related to tolerance.

Individual responses to β2 agonists can vary considerably due to factors such as genetic polymorphisms and underlying inflammation. Improper use or overuse of these agents may lead to rapid tolerance, diminishing their bronchodilator effectiveness during critical exacerbations.[67] As β2 agonists are often prescribed as part of inhaled β2 agonist–corticosteroid combinations for first-line **asthma** maintenance therapy, patients must receive thorough education on proper inhaler technique, dosing, and timing to maximize efficacy and minimize the risk of adverse events.

The optimal care of patients with chronic respiratory inflammatory diseases requires a coordinated, interprofessional approach. Pulmonologists, allergists, primary care physicians, pharmacists, and nurses must work collaboratively to select and customize therapies, tailoring treatment regimens to individual patient-specific factors and disease severity. This includes tailoring treatment regimens, monitoring efficacy and safety, regularly evaluating lung function, and assessing for adverse effects such as cardiovascular complications. Effective communication among healthcare team members is essential to ensure timely updates on patient status and appropriate adjustments to treatment plans.

As β2 agonists may exert systemic effects, such as influencing heart rate and interacting with other medications, clinicians in related specialties, including cardiology and obstetrics, must also remain informed about the use of these **drugs**. In certain cases, adjustments in management may be necessary to prevent exacerbation of comorbid conditions or triggering adverse drug interactions.

Finally, public education is crucial. Patients and caregivers should be informed of the risks associated with inappropriate or unsupervised use of β2 agonists (eg, misuse for presumed anabolic purposes). Concurrently, emergency physicians and nurses require specific training to recognize and manage β2 agonist overdose, ensuring timely and effective intervention.

Review Questions

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Tables

Table 1. Routes of Administration of the Most Commonly Used Beta-2 Agonists

Class	Drugs	Available Route(s) of Administration
Short-acting β2 agonists	Albuterol (Salbutamol)	Inhaler (MDI and DPI), nebulizer, oral, IV, subcutaneous, and intramuscular [23][24][25]
	Levalbuterol	Inhaler (MDI) and nebulizer [23][26][27]
	Terbutaline	Inhaler, oral, IV, and subcutaneous [8][28]
	Metaproterenol	Inhaler, nebulizer, and oral [29][30]
Long-acting β2 agonists	Salmeterol	Inhaler (MDI and DPI) [31][32]
	Formoterol	Inhaler (MDI and DPI), nebulizer, and oral [33][34][35]
Ultra–long-acting β2 agonists	Olodaterol	Inhaler [36]
	Vilanterol	Inhaler [37]
	Indacaterol	Inhaler [38]

Table 2. Routes of Administration of Some Popular Beta Blockers

Beta Antagonist Drug	Available Route(s) of Administration
Propranolol	Oral and intravenous
Timolol	Oral, topical (ophthalmic and cutaneous), and sublingual
Carvedilol	Oral, transdermal, and buccal [44][45]

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